

ISHA KHAN

Rochester, NY

toishakhan@gmail.com – 203-892-3084 – LinkedIn – Web page – GitHub

Research Interests: Dynamic Macroeconomics; Monetary Economics; Inflation Dynamics; International Macroeconomics; Structural and Computational Macroeconomics

RESEARCH

Do Trade Deficits Really De-Industrialize? Manufacturing Shares and Trade Balances

Co-authored with Christopher Ball, Ph.D. – Forthcoming, Economics Letters

Empirical analysis of trade balances and manufacturing outcomes using country-level panel data. Examines whether persistent trade deficits systematically predict manufacturing decline across countries.

- Conducted all econometrics work for this paper, including implementing panel regression models with Newey-West and clustered standard errors.
- Wrote custom R/Python code to construct a country-level panel using OECD, World Bank, and other global macro sources.
- Performed multiple robustness tests to validate the findings, which challenge a common policy assumption: trade deficits alone do not systematically predict manufacturing decline.

150 Years of Inflation: Monetary, New Keynesian, and Fiscal Theory Conceptualizations

Research Assistant to Christopher Ball, Ph.D. – In Progress

Researched relationships between long-run inflation patterns and theoretical models (Monetarist, New Keynesian, and Fiscal Theory) by translating conceptual frameworks into empirical code logic. Using more than 150 years of global data and modern back-testing methods, the project tests and compares the empirical validity of prevailing theories of inflation.

- Worked with datasets spanning more than 150 years and 100+ countries.
- Conducted time-series and panel econometric analysis, along with long-horizon comparisons.
- Developed Markov-chain transition models to estimate long-run inflation regime probabilities and characterize monetary regime persistence.
- Translated the theoretical frameworks of Monetarist, New Keynesian, and Fiscal Theory models into empirically testable code (R).

EDUCATION

University of Rochester, Rochester, NY

2026 – Present

Ph.D. in Economics

Quinnipiac University, Hamden, CT

May 2026

B.S. Economics, B.A. Mathematics, B.S. Data Science (Honors)

Cumulative GPA: 3.96

GRE: 170Q | 157V | 4.0 AW

Honors/Awards: National Merit Commended Scholar; Dean's List Fall 2023/Fall 2025; Dean's List with Honors Spring 2024/Fall 2024/Spring 2025

DATA SCIENCE COURSEWORK

Algorithms of Data Science (Proof-Based) – eigendecomposition, MLE, neural networks, steepest descent
Machine Learning – ridge/lasso regression, bootstrapping, random forests
Data Mining – data visualizations, heatmapping, data cleaning
Advanced Applied Statistics – regularization, ANOVA, model diagnostics

MATH COURSEWORK

Mathematical Economics (PhD Prep) – Ramsey growth; money-in-utility and cash-in-advance models; discrete-time dynamic programming (Bellman); continuous-time optimal control (Hamiltonians); phase diagrams; numerical simulation.
Real Analysis – continuity, compactness, convergence, ε - δ proofs
Advanced (Proof-Based) Calculus – differentiation in R^n , implicit function theorem, multivariable optimization
Math Stats and Probability I/II – distributions, continuous/discrete expectations, Bayesian statistics
Multivariate Calculus (Calc III) – vector calculus, line/surface integrals, partial differentiation
Applied Stats Theory – hypothesis testing
Linear Algebra – eigenvalues/eigenvectors, matrix decomposition (LU, QR, SVD)
Foundations of Advanced Mathematics/Proof Writing – equivalence relations, functions, sets, logic, proof techniques

ECONOMICS COURSEWORK

Econometrics – OLS, inference, regression diagnostics
Advanced Econometrics – panel data models, IV, clustered SEs, cross-validation
Game Theory – Nash equilibrium, backward induction, mixed strategies
Money, Banking, and Finance – monetary policy, clearing house operations, intraday trading, federal funds market
Intermediate Macroeconomics – DSGE foundations, Phillips curve, Okun’s law, New Keynesian, Monetarist, Fiscal Theory of the Price Level
Intermediate Microeconomics – optimization, firm behavior, market types, consumer theory

TECHNICAL SKILLS

Statistical/Quantitative Methods:

Dynamic Programming (Bellman Equations), Linear/Logistic Regression, PCR, PCA, Ridge/Lasso, K-Means Clustering, Random Forests, Mean-Variance Optimization

Programming & Tools:

Python (NumPy, pandas, scikit-learn, matplotlib); R; SQL; Stata; LaTeX; Jupyter Notebook; Microsoft Excel (Financial Modeling, Pivot Tables)

TEACHING EXPERIENCE

Calculus 2 Peer Fellow

Quinnipiac University, Hamden, CT

- Run review sessions to cover difficult topics.
- Meet with students individually when extra help is needed.

Computer Information Systems Engagement Fellow

Quinnipiac University, Hamden, CT

- Support the professor during class sessions and in maintaining course page online.

- Manage and proctor exams/quizzes.

SAT Math Tutor

Schoolhouse, Online

- Certified by Schoolhouse to tutor both SAT Math and SAT Reading and Writing.
- Led sessions with groups of 15–20 students, covering a range of SAT Math topics and providing one-on-one support when needed.
- Received 92% ‘Helpful Ratings’, with 42% rating sessions as ‘Super Helpful’.

LEADERSHIP

Quinnipiac Economics Club President – Active Member Since Fall 2023; President Since Summer 2025

Investment Club – Active Member Since Fall 2023

Quinnipiac Economics & Data Science – Student Influencer Since Spring 2025

EXPERIENCE

Investments Intern

Avangrid, Orange, CT

June 2025 – August 2025

- Improved revised budget input efficiency by over 60%, reducing average input time from ~14 minutes to less than 5 through an integrated Excel dataflow structure.
- Participated in financial analysis projects, including capital budgeting and scenario forecasting, collaborating closely with senior investment analysts in producing reports.
- Involved with 2 pilot projects due to domain knowledge in capital expenditure processing and data science.

Senior Researcher

Quinnipiac Global Economics Team, Hamden, CT

Fall 2024 - Spring 2026

- Simplifies complex economic ideas and terms to make compact yet coherent reports that attempt to analyze four economic subjects (countries, commodities, currencies, etc.) per report.
- Developed data visualizations in collaboration with other coders to enhance the readability of the reports.
- Questioned a prevailing assumption in a weekly report and rewrote the structure to improve consistency and usability while increasing output by 25%.